
SSM Optimization

Weekly Meeting

10/04/26

Results on Wikitext2 – Quamba2

On Wikitext2

Model	Quamba2							SINQ	
	FP16	W8A8		W4A8		W4A16		W4A16	
	PPL	PPL	KLD	PPL	KLD	PPL	KLD	PPL	KLD
Mamba-2.8b	9.452	9.997	5.795e-2	10.200	8.084e-2	9.718	3.245e-2	9.714	2.857e-2
Mamba-1.4b	10.747	11.499	6.582e-2	11.822	9.345e-2	11.156	3.610e-2	11.078	3.086e-2
Mamba-790m	11.956	19.707	4.981e-1	20.074	5.180e-1	12.792	6.814e-2	1.524e6	11.748



- Comparing W4A16 settings, SINQ achieves slightly better results (both in terms of perplexity and KL divergence) for Mamba-1.4b and -2.8b
- Mamba-790m is very unstable

Model	SINQ	
	W4A16	
	PPL	KLD
Mamba-790m	1.524e6	11.748

Baselines

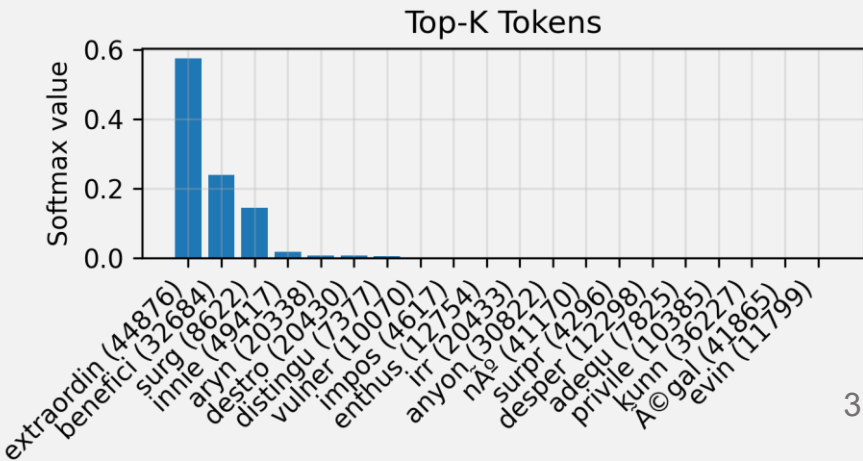
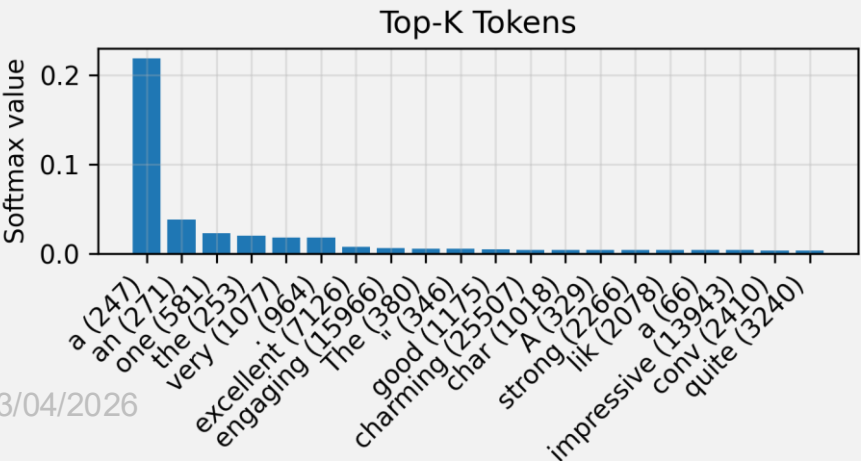
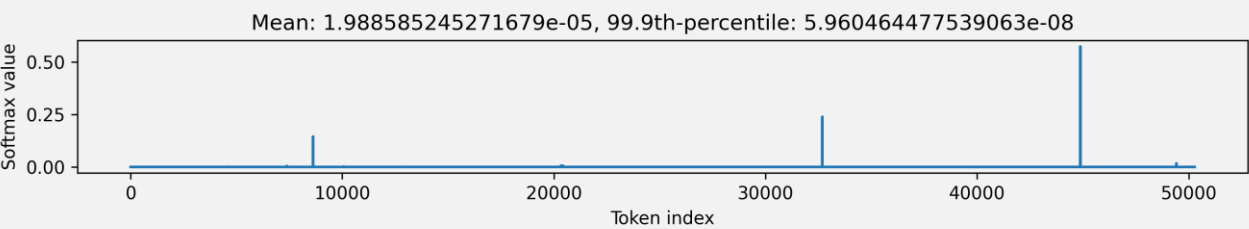
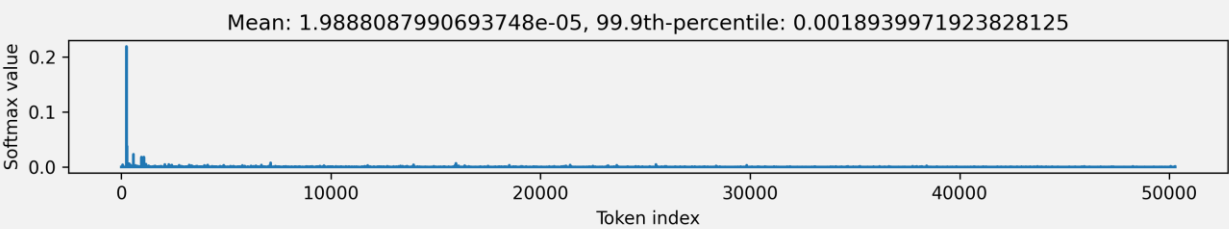
Results on Wikitext2 – SINQ

CONTEXT: [...] In 2000 Boulter had a guest @-@ starring role on the television series The Bill ; he portrayed " Scott Parry " in the episode, " In Safe Hands ". Boulter starred as " Scott " in the play Herons written by Simon Stephens, which was performed in 2001 at the Royal Court Theatre. A review of Boulter's performance in The Independent on Sunday described him as "

TRUE: hor | **FP:** a | **Q:** extraordin

FP model

Q model



Model	SINQ	
	W4A16	
	PPL	KLD
Mamba-790m	1.524e6	11.748

Baselines

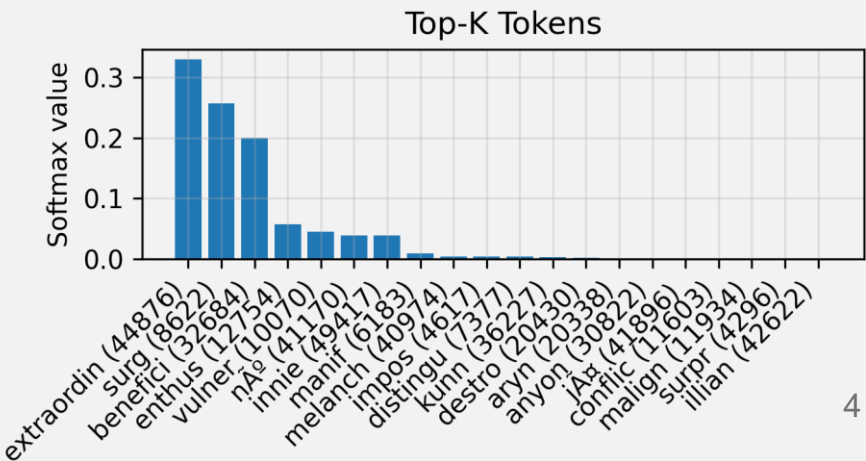
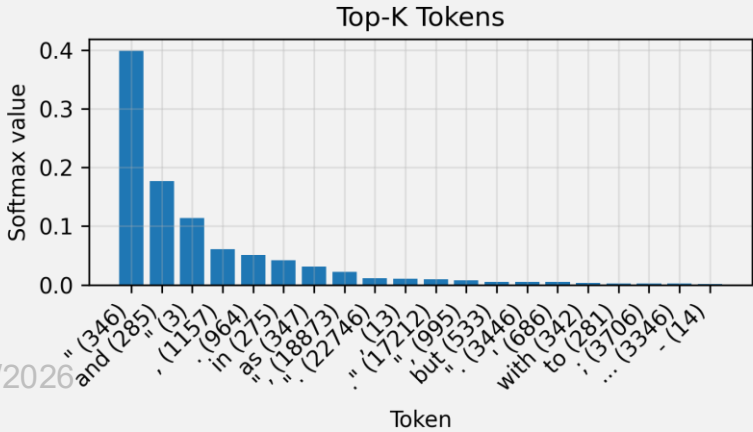
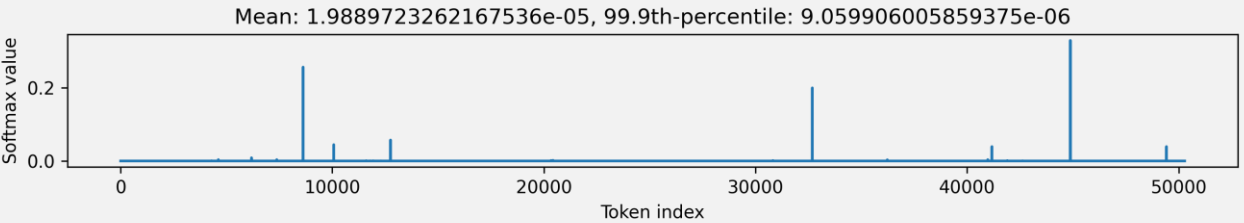
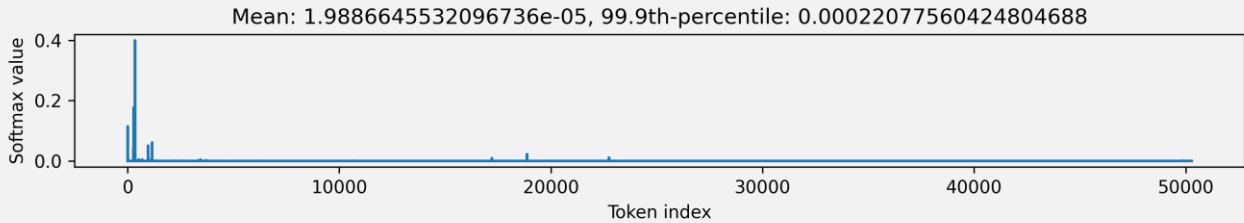
Results on Wikitext2 – SINQ

CONTEXT: [...] In 2000 Boulter had a guest @-@ starring role on the television series The Bill ; he portrayed " Scott Parry " in the episode, " In Safe Hands ". Boulter starred as " Scott " in the play Herons written by Simon Stephens, which was performed in 2001 at the Royal Court Theatre. A review of Boulter's performance in The Independent on Sunday described him as " horribly menacing"

TRUE: " | **FP:** " | **Q:** extraordin

FP model

Q model



Model	SINQ	
	W4A16	
	PPL	KLD
Mamba-790m	1.524e6	11.748

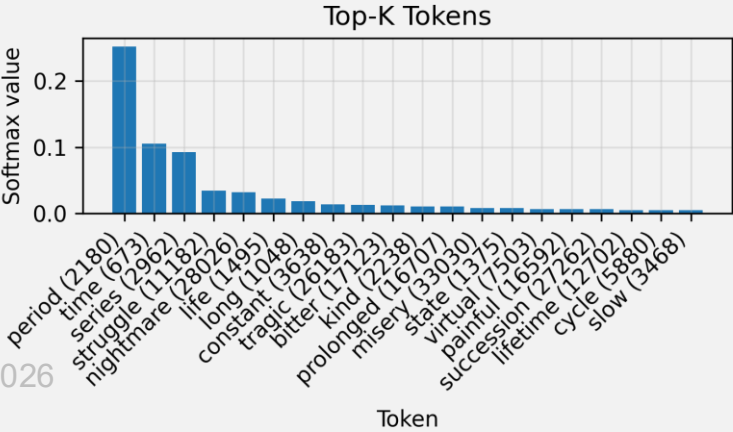
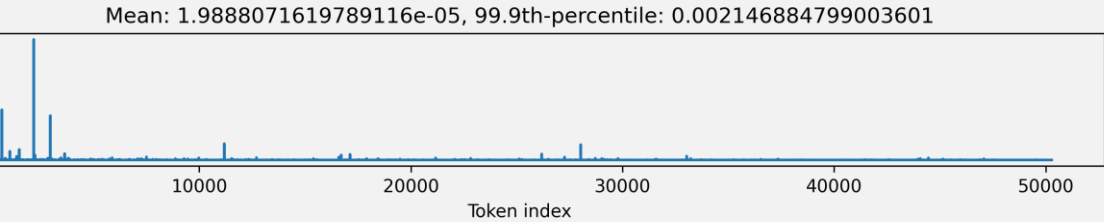
Baselines

Results on Wikitext2 – SINQ

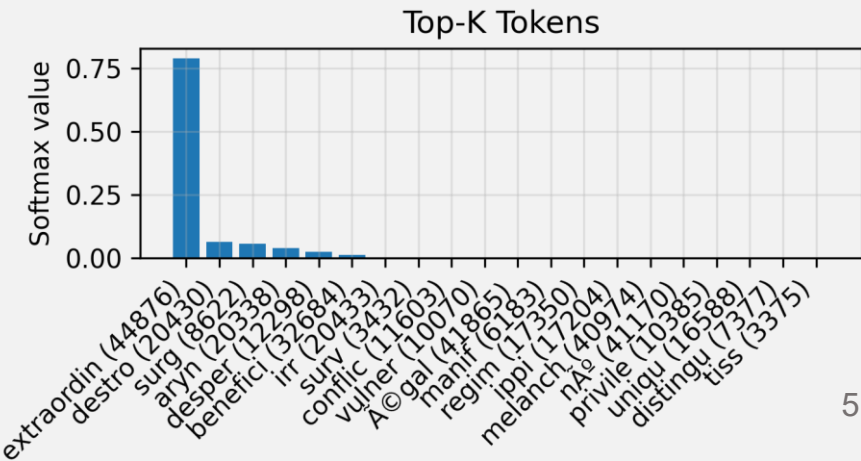
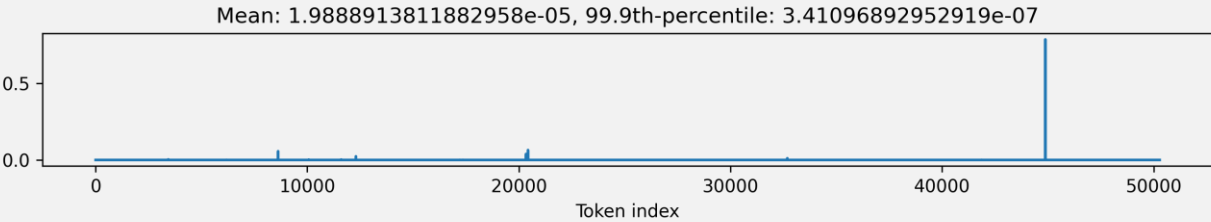
CONTEXT: [...] == Television == \n\n\n\n\n\n\n == Theatre == \n\n\n\n\n\n\n\n\n\n\n\n = Du Fu = \n\n\n\n\n\n\n Du Fu (Wade – Giles : Tu Fu ; Chinese : 杜甫 ; 712 – 770) was a prominent Chinese poet of the Tang dynasty. Along with Li Bai (Li Po), he is frequently called the greatest of the Chinese poets. His greatest ambition was to serve his country as a successful civil servant, but he proved unable to make the necessary accommodations. His life, like the whole country, was devastated by the An Lushan Rebellion of 755, and his last 15 years were a"

TRUE: time | **FP:** period | **Q:** extraordin

FP model



Q model



Model	SINQ	
	W4A16	
	PPL	KLD
Mamba-790m	1.524e6	11.748

Baselines

Results on Wikitext2 – SINQ

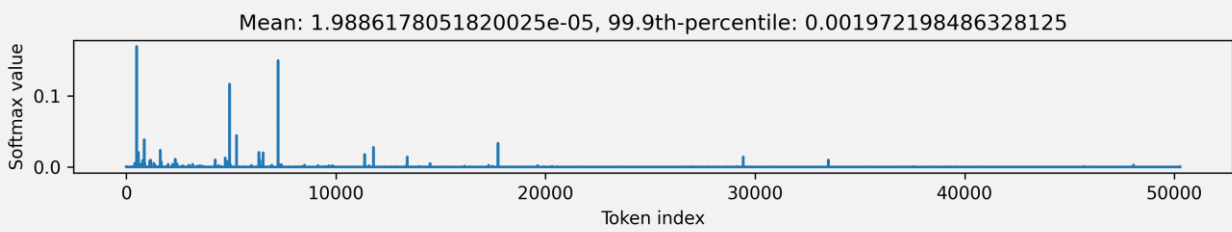
CONTEXT: [...] In 762, he left the city to escape a rebellion, but he returned in summer 764 when he was appointed an advisor to Yan, who was involved in campaigns against the Tibetan Empire.

=== Last years ===

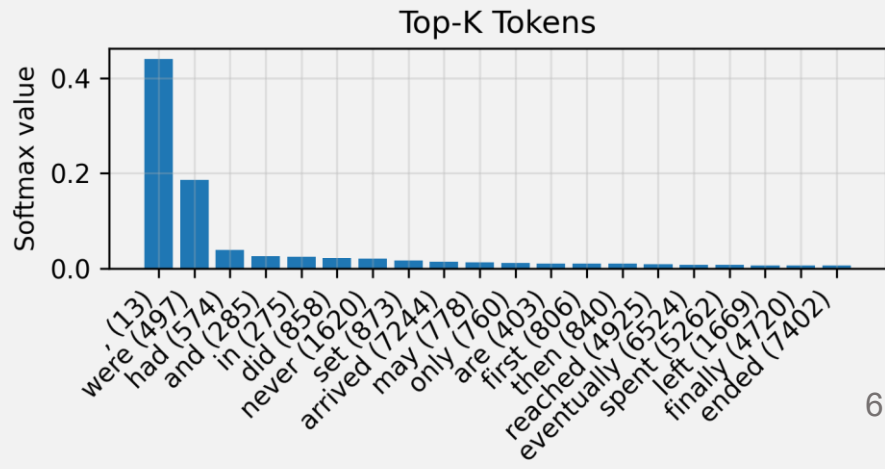
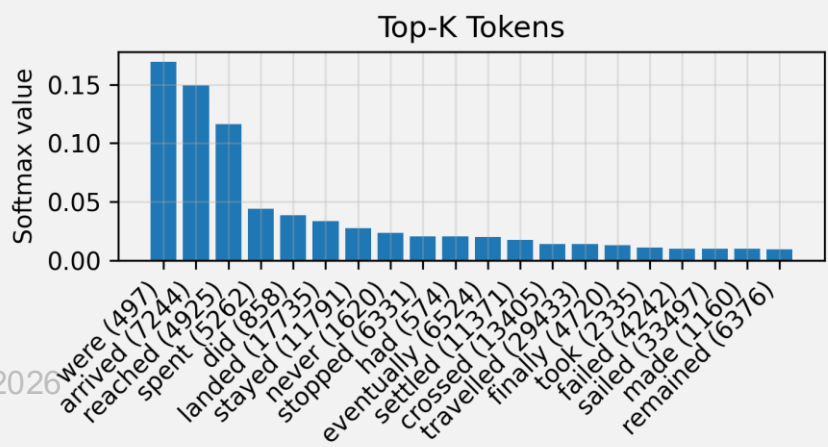
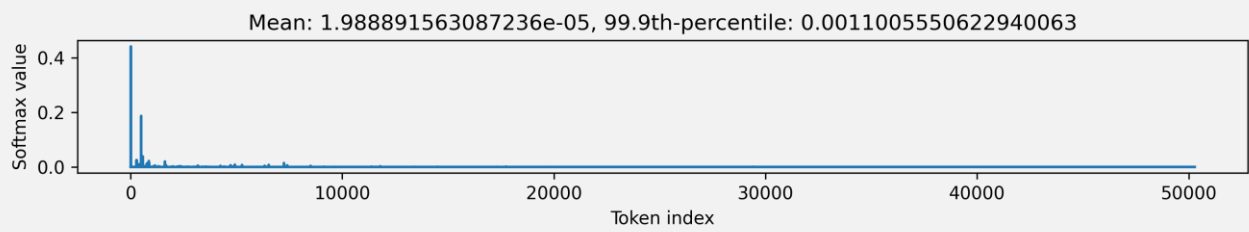
Luoyang, the region of his birthplace, was recovered by government forces in the winter of 762, and in the spring of 765 Du Fu and his family sailed down the Yangtze, apparently with the intention of making their way there. They"

TRUE: traveled | **FP:** were | **Q:** ,

FP model



Q model



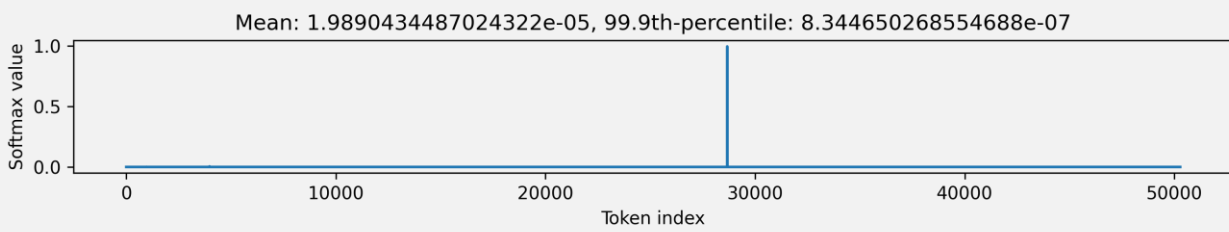
Model	SINQ	
	W4A16	
	PPL	KLD
Mamba-790m	1.524e6	11.748

Baselines

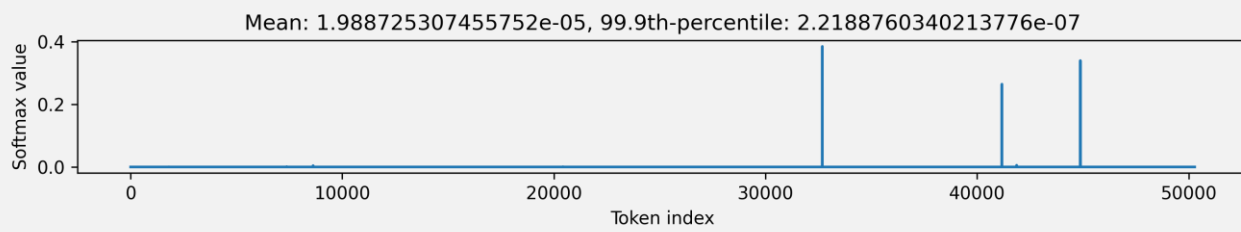
Results on Wikitext2 – SINQ

CONTEXT: [...] The sisters were refitted in late 1940 in preparation for war, which included the fitting of external degaussing coils and additional AA directors.
=== World War II ===
When Japan began the Pacific War on 8 December, the sisters sortied for the Bonin Islands with four other battleships and the light carrier Hōshō as distant cover for the fleet attacking Pearl"
TRUE: Harbor | **FP:** Harbor | **Q:** benefici,

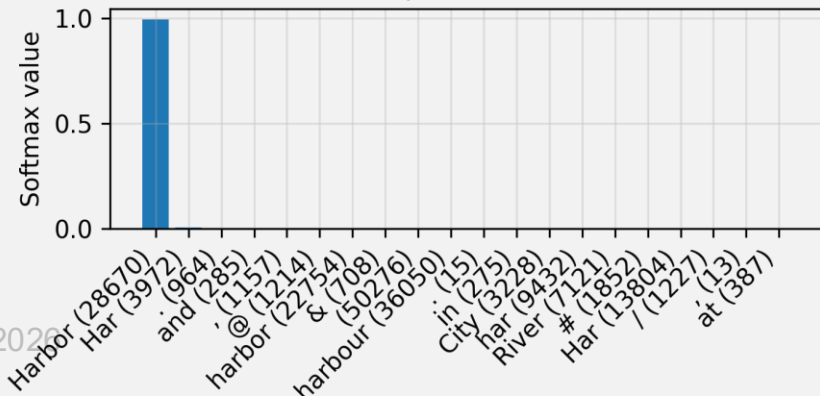
FP model



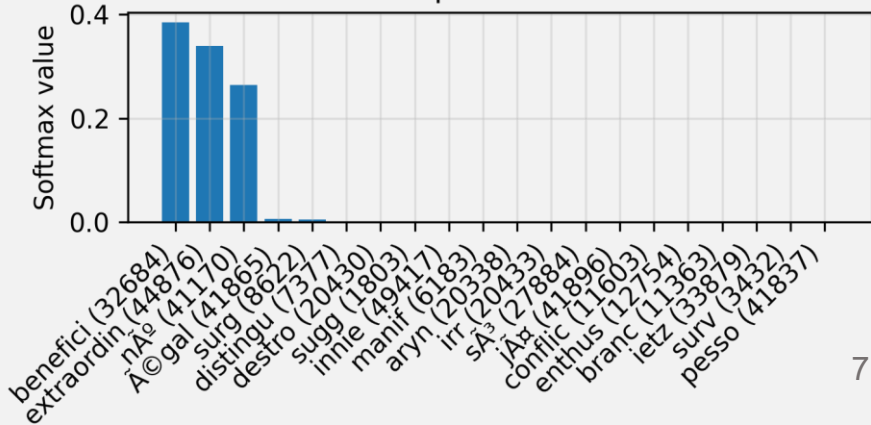
Q model



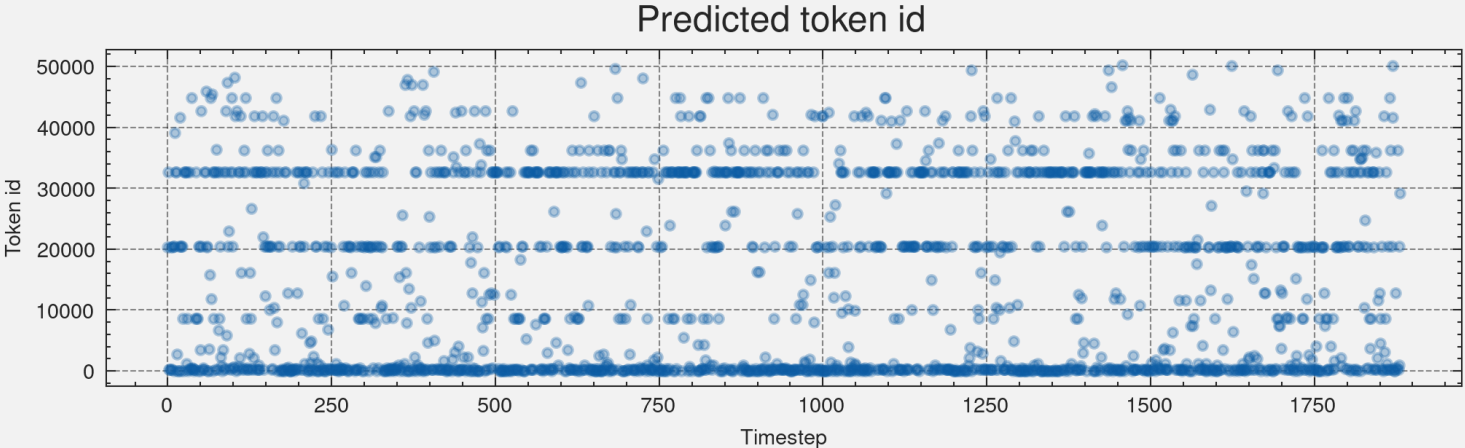
Top-K Tokens



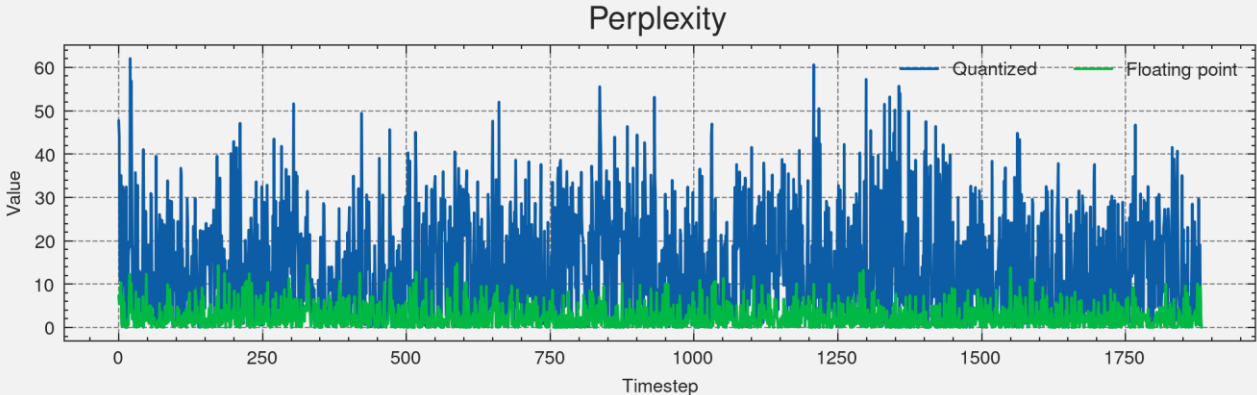
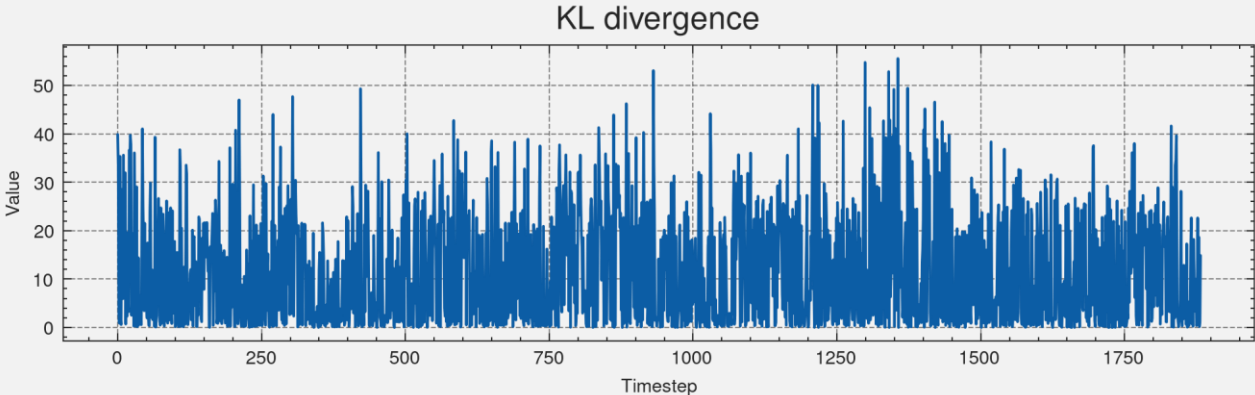
Top-K Tokens



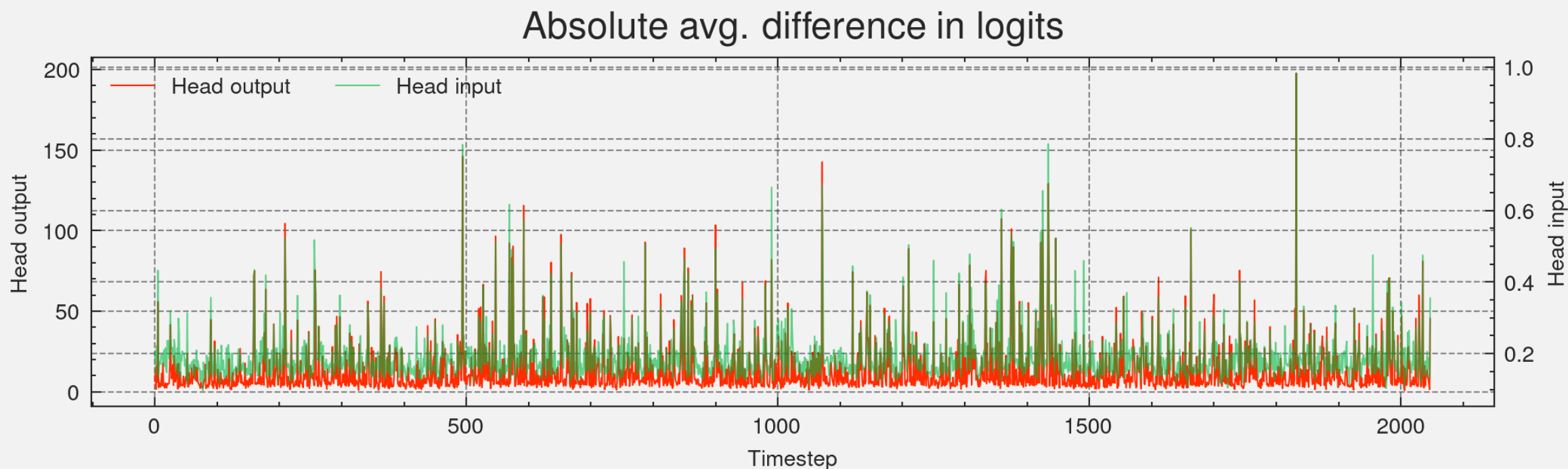
Results on Wikitext2 – SINQ



```
{ 'benefici': 382,  
  'destro': 195,  
  ',': 153,  
  'the': 119,  
  'and': 119,  
  'aryn': 89,  
  'surg': 69,  
  '-': 68,  
  'kunn': 68,  
  '': 36,  
  'The': 25,  
  'extraordin': 24,  
  'in': 24,  
  'I': 24,  
  'Aegal': 23,  
  'of': 20,  
  'to': 20,  
  '@': 18,  
  'a': 18,  
  '(': 17,  
  'nÃ': 16,  
  'were': 15,  
  '': 15,  
  '\n': 13,  
  'jÃ': 13,  
  ...
```



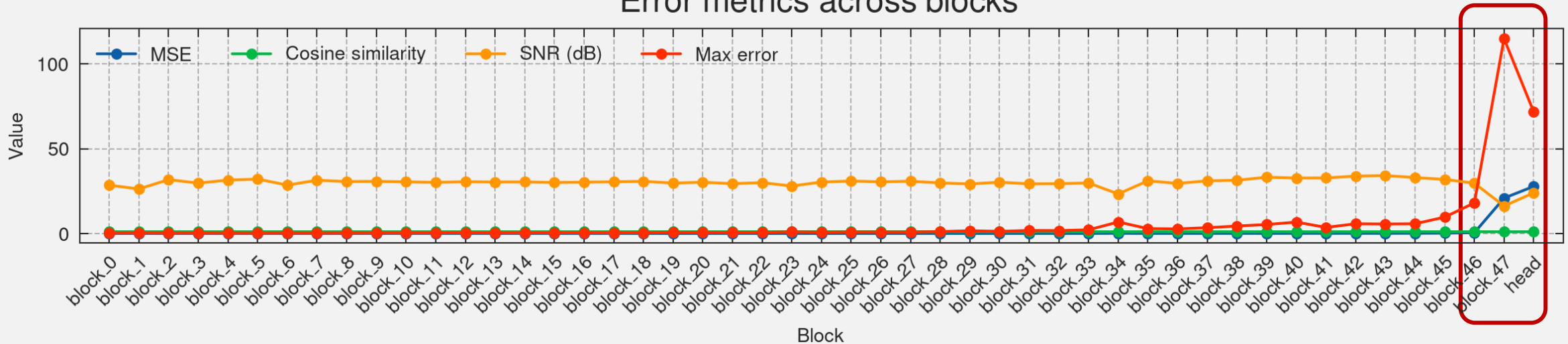
Results on Wikitext2 – SINQ



Results on Wikitext2 – SINQ – M.-790m

Analyzed the output error per block when given always FP inputs:

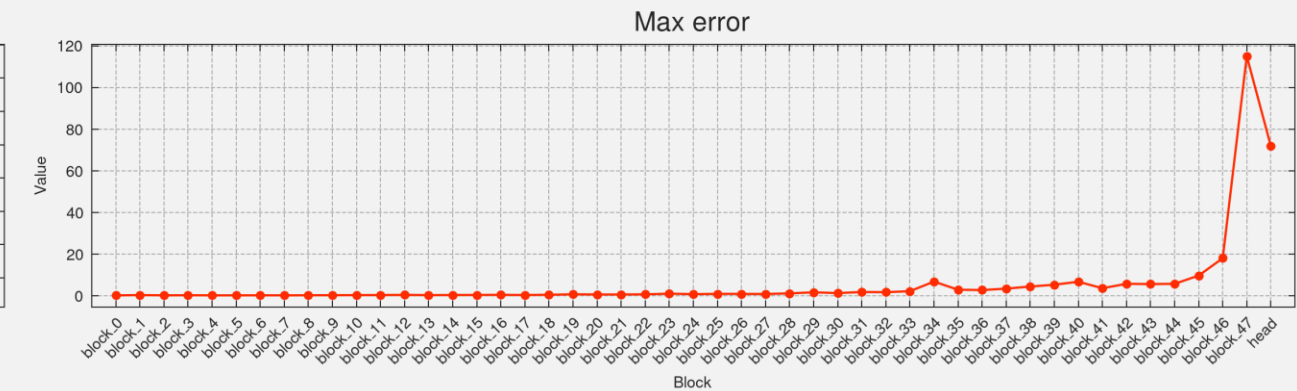
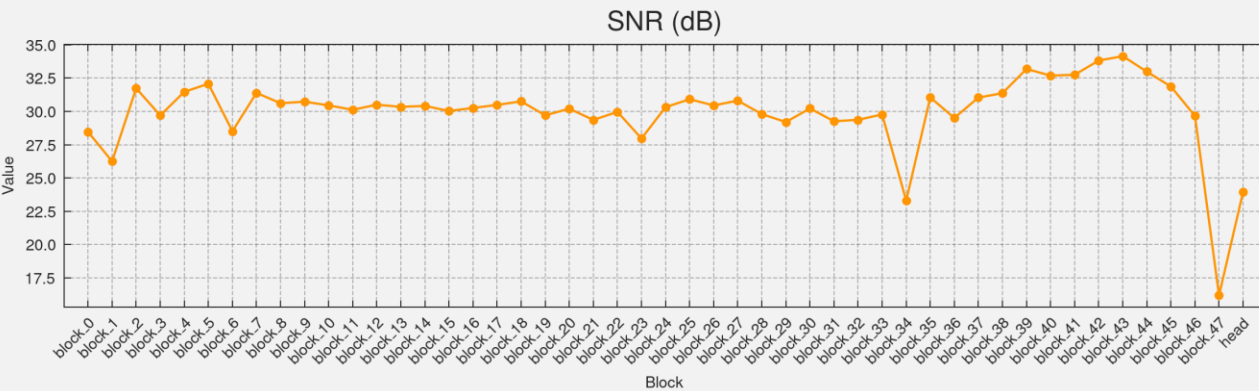
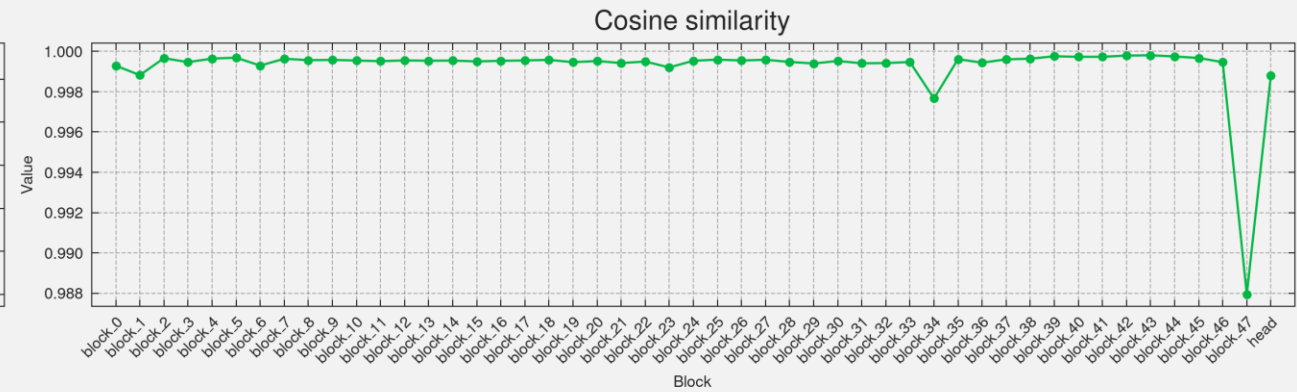
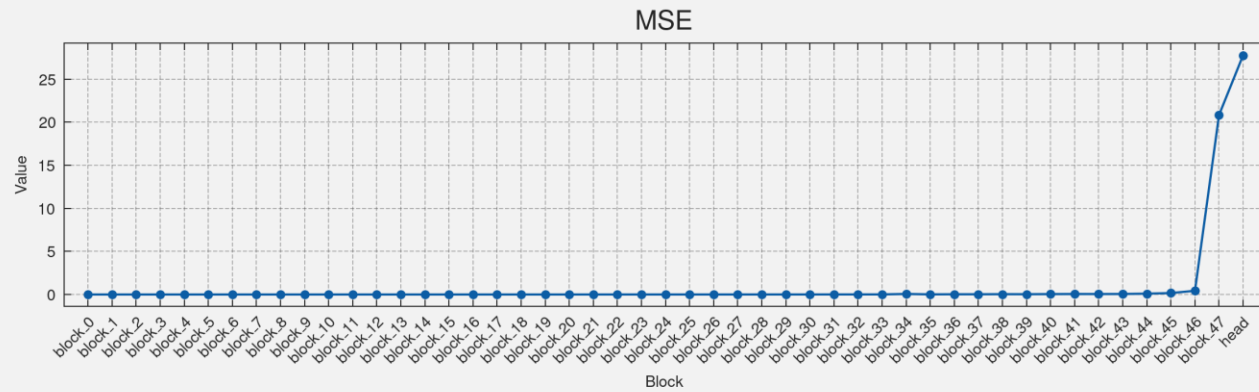
Error metrics across blocks



- Output quality starts degrading in blocks 46 and even more 47
- Interestingly, by completely skipping block 47 the output token is even the correct one

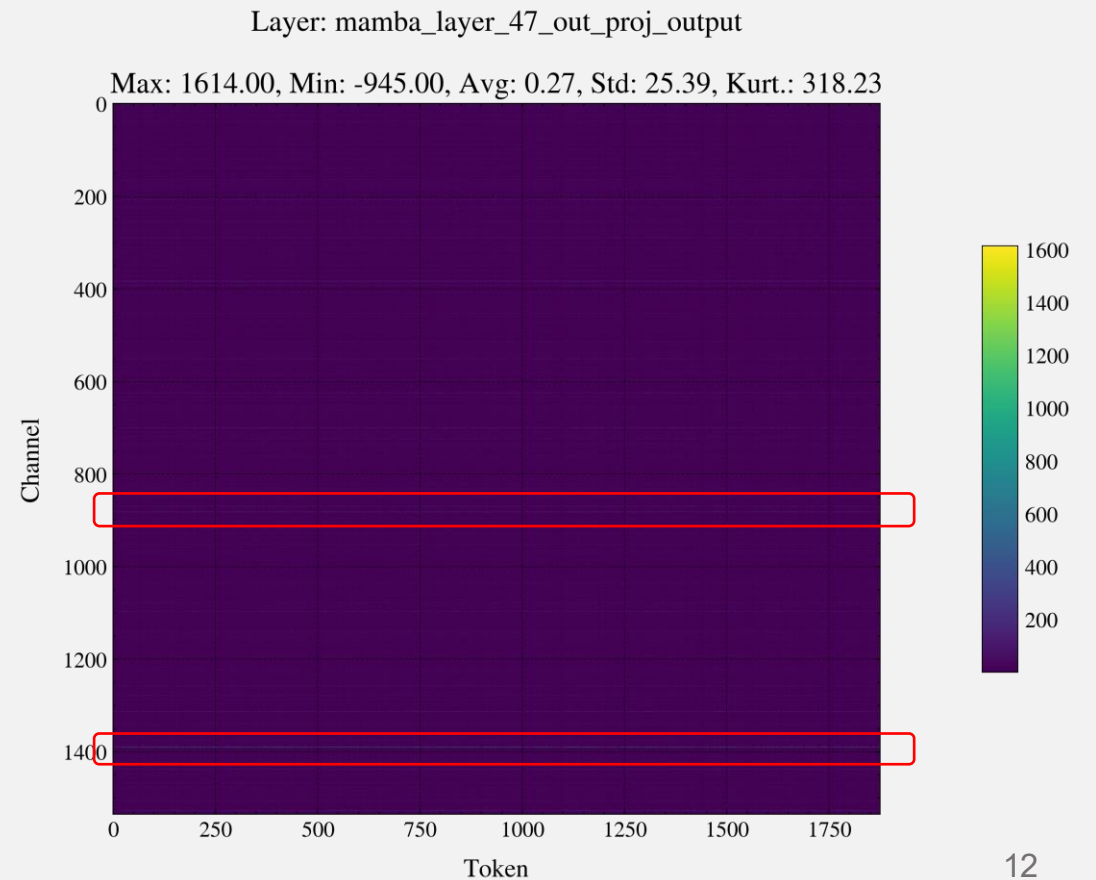
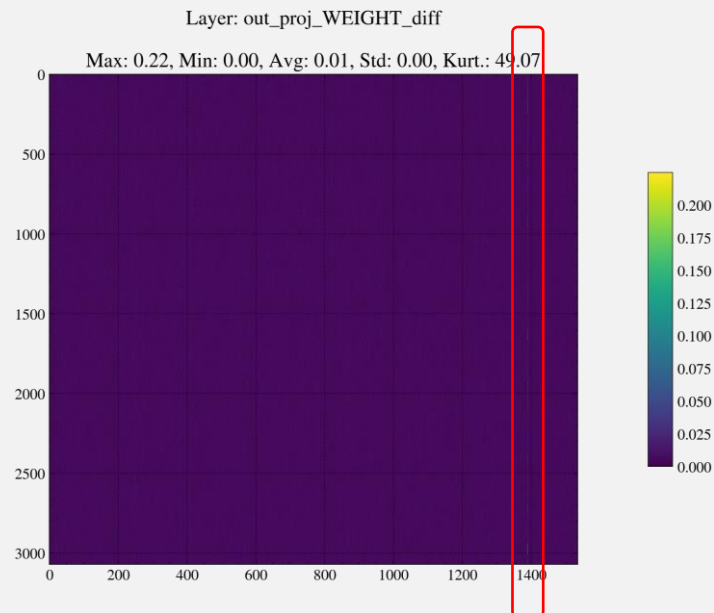
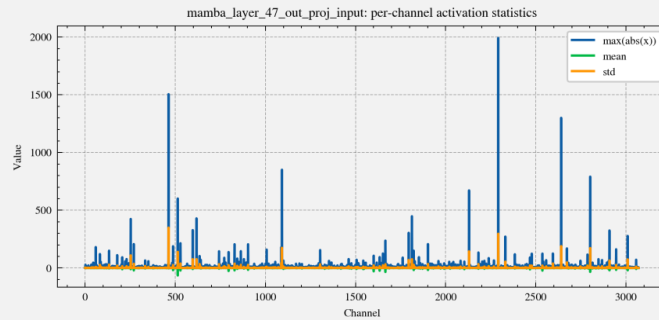
Baselines

Results on Wikitext2 – SINQ – M.-790m

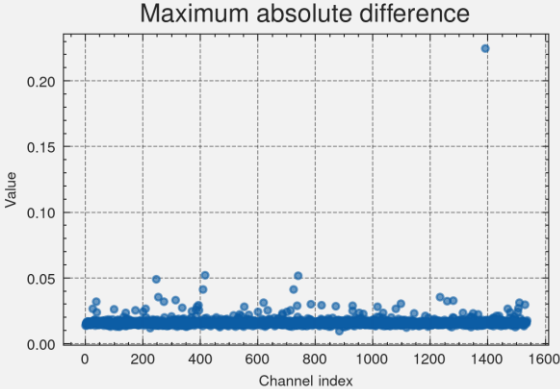
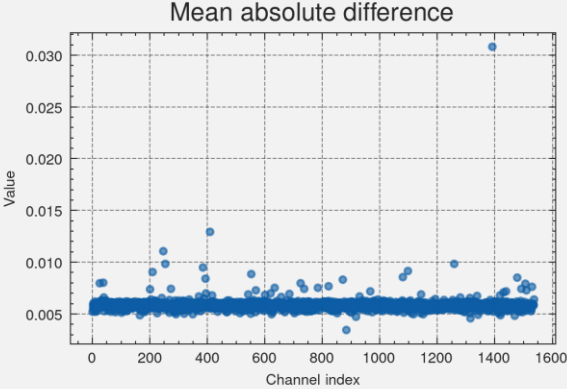
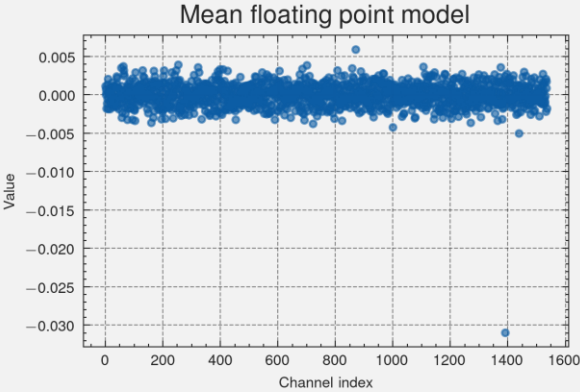
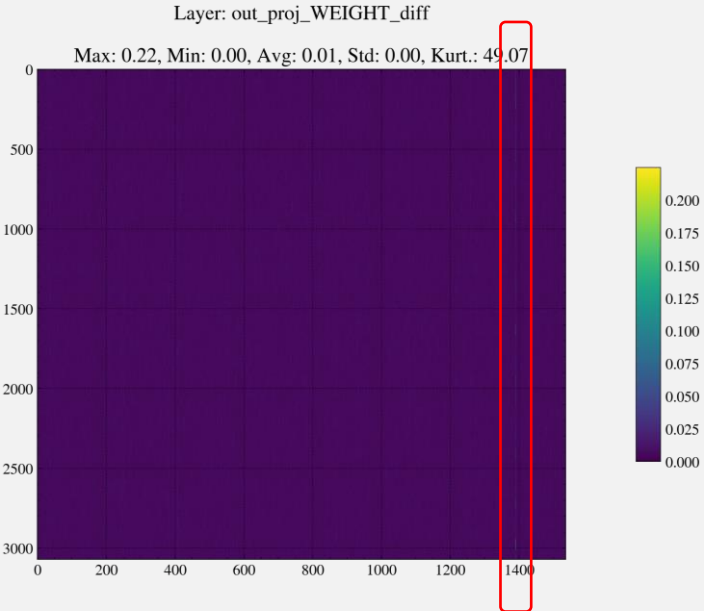


Activations at block 47

FP model's activations passed to each block



Weights: out_proj@47



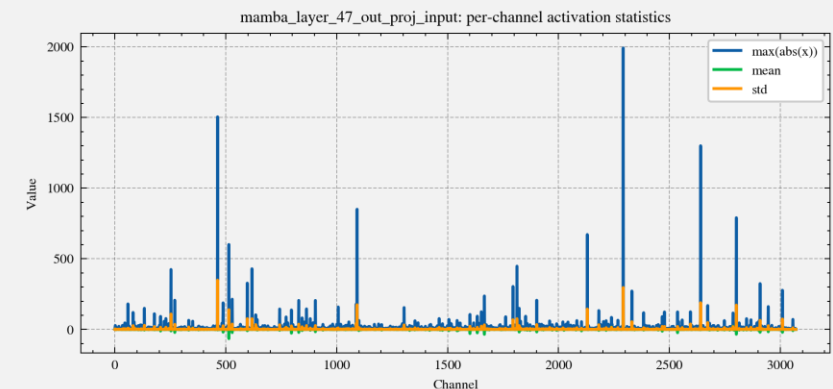
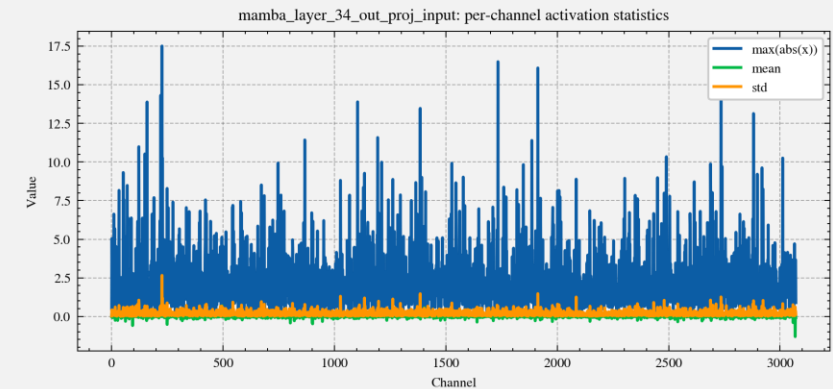
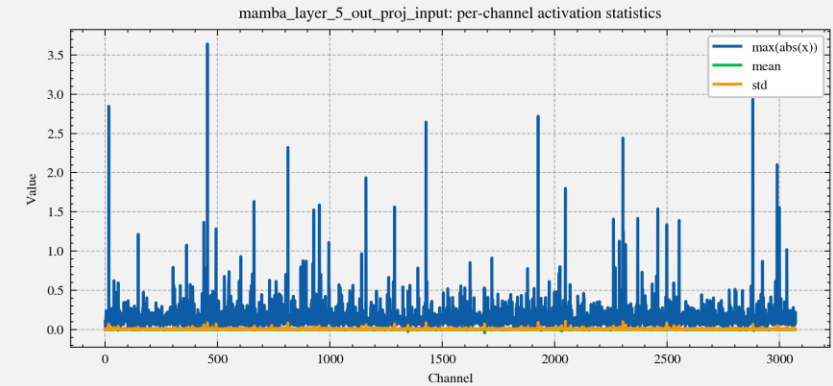
Baselines

Max. abs. (avg.) FP weight of out_proj:

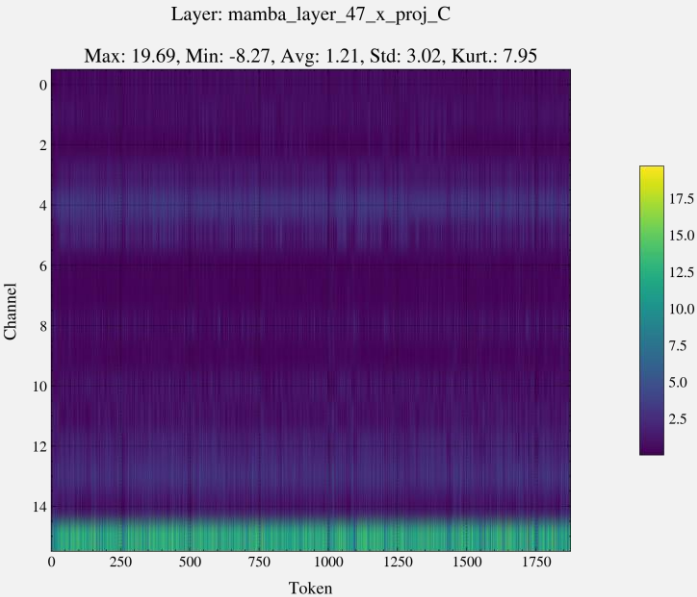
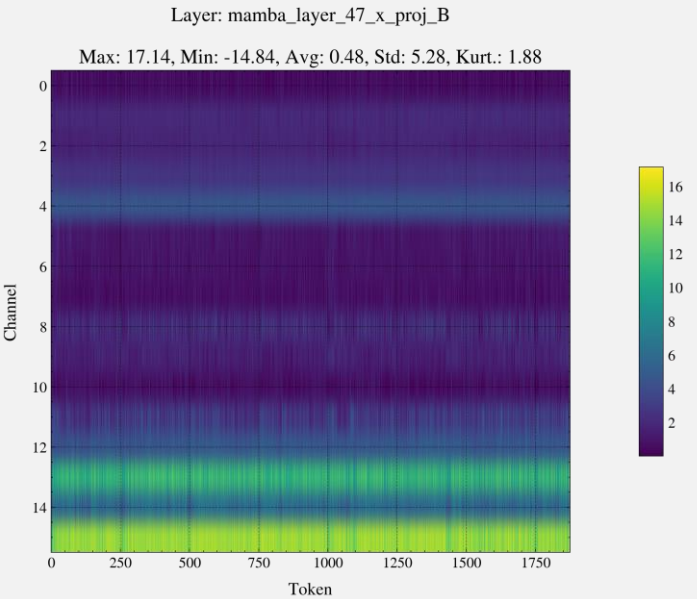
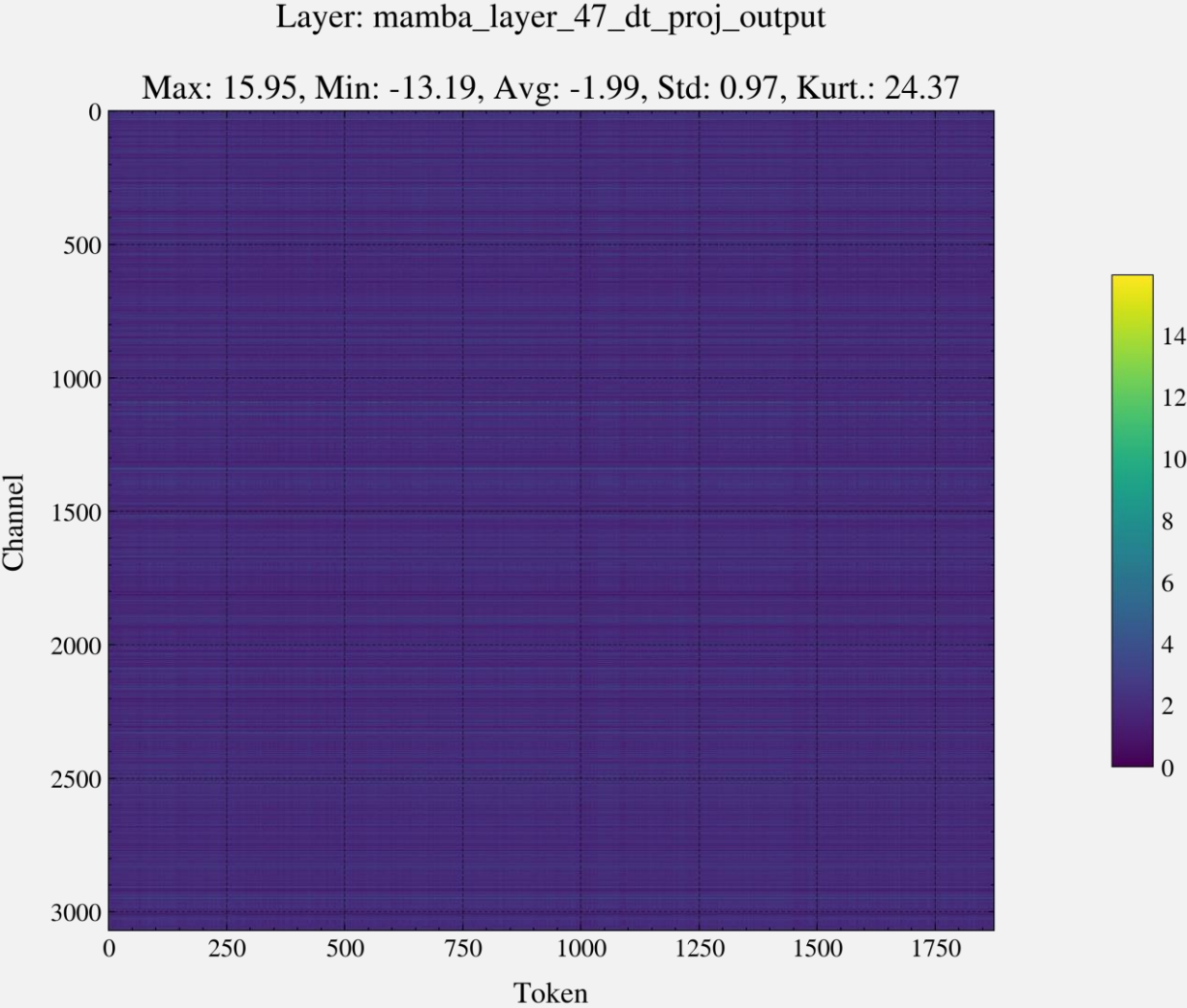
- Block 5: 1.67 (0.00)
- Block 34: 1.96 (0.00)
- Block 47: 3.81 (0.00)

Max. (avg.) abs. error at block 47:

- In_proj: 0.04 (0.01)
- X_proj: 0.06 (0.01)
- Dt_proj: 0.05 (0.00)
- Out_proj: 0.22 (0.01)



Baselines



Results on LAMBADA – Quamba2

Compared results of my runs with the numbers reported in the paper to verify reproducibility (possible only on LAMBADA)

On LAMBADA								
Model	Quamba2							
	FP16		W8A8		W4A8		W4A16	
	Mine	Paper	Mine	Paper	Mine	Paper	Mine	Paper
Mamba-2.8b	69.1	69.1	65.9 (-0.2)	65.7	63.4 (-0.1)	63.5	65.2 (-0.8)	66.0
Mamba-1.4b	64.9	64.9	62.2 (-0.1)	62.3	61.8 (+0.3)	61.5	64.4 (+0.8)	63.6

Mamba-3

- New **SSM discretization technique** replacing the zero-order hold technique (used for Mamba-1 and -2), based on the trapezoidal rule.
 - A, B and x are constants between time-steps → convex combination between the segment endpoints
 - Blend of current and previous inputs is used to update the state
- Complex-valued state update rule to mimic the rotation for the state update
 - This improves state-tracking performance
- Introduce rank dimension on B (that goes from N to NxR) to increase the representational capacity and arithmetic intensity of the model
 - Similar decoding latency (~1.15x increase)
 - Higher training cost

Next steps

- As we discussed:
 - Act. quantization
 - Experiments with hybrid architectures

Extra:

- Other paper?